Operations Research

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Solution to the Demo Exercises, Sheet 3

Demo Exercises

Exercise D3. 1. *Revised Simplex*

Problem

Solve the following LP using the revised simplex algorithm. Start with the feasible basis (x_1, x_6).

$$\begin{aligned} \underset{\text{s.t.}}{\text{Max}}\ & 12x_1 + 20x_2 + 16x_3 + 12x_4 \\ 4x_1 + 8x_2 + 2x_3 &\leq 400 \\ 2x_1 + 5x_2 + 10x_3 + 4x_4 &\leq 400 \\ x &\geq 0 \end{aligned}$$

Solution

We first bring the LP into canonical form by introducing the slack variables $x_5, x_6 \geq 0$ and obtain

$$A = \begin{pmatrix} 4 & 8 & 2 & 0 & 1 & 0 \\ 2 & 5 & 10 & 4 & 0 & 1 \end{pmatrix} \; b = \begin{pmatrix} 400 \\ 400 \end{pmatrix} \; c = (12, 20, 16, 12, 0, 0)^{\tau}$$

Iteration 1 Basic variables: $x_B = (x_1, x_6)^T$, Nonbasic variables: $x_N = (x_2, x_3, x_4, x_5)^T$

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix} \; N = \begin{pmatrix} 8 & 2 & 0 & 1 \\ 5 & 10 & 4 & 0 \end{pmatrix} \; c_B = \begin{pmatrix} 12 \\ 0 \end{pmatrix} \; c_N = (20, 16, 12, 0)^T$$

$$B^{-1} = \begin{pmatrix} 1/4 & 0 \\ -1/2 & 1 \end{pmatrix} \; N' = B^{-1}N = \begin{pmatrix} 2 & 1/2 & 0 & 1/4 \\ 1 & 9 & 4 & -1/2 \end{pmatrix}$$

$$c'_N = -(c_N - N'^T c_B) = (4, -10, \mathbf{-12}, 3)^T, \; x_B = b' = B^{-1}b = (100, 200)^T$$

We select the x_4 column as the pivot column and determine the quotients from the right-hand side (b_j) with the coefficients of the pivot column, which are positive. We therefore only obtain a quotient for the second row. The second row is our pivot row and we carry out the basis-exchange: $x_6 \leftrightarrow x_4$.

Iteration 2 $x_B = (x_1, x_4)^T$, $x_N = (x_2, x_3, x_5, x_6)^T$

$$B = \begin{pmatrix} 4 & 0 \\ 2 & 4 \end{pmatrix} \quad N = \begin{pmatrix} 8 & 2 & 1 & 0 \\ 5 & 10 & 0 & 1 \end{pmatrix} \quad c_B = (12, 12)^T,\ c_N = (20, 16, 0, 0)^T$$

$$B^{-1} = \begin{pmatrix} 1/4 & 0 \\ -1/8 & 1/4 \end{pmatrix} \quad N' = \begin{pmatrix} 2 & 1/2 & 1/4 & 0 \\ 1/4 & 9/4 & -1/8 & 1/4 \end{pmatrix}$$

$$c'_N = (7, 17, 3/2, 3)^T,\ x_B = b' = (100, 50)^T$$

The stopping criterion is fulfilled (only positive entries in c'_N), so $x = (100, 0, 0, 50, 0, 0)^T$ with $Z = c_B^T b' = 1800$ is the optimal solution to the problem.

Exercise D3. 2. *Waldgeist Distillery*

Problem

The Waldgeist distillery, led by Steffen, produces two different spirits from raspberries, blueberries and strawberries: the Himmelrot liqueur and the Nachtblau fruit brandy. The available capacities, the required quantity of raspberries and blueberries in kg as well as the unit price and costs per bottle of each variety can be seen in the following table:

Resources	Himmelrot	Nachtblau	Capacity
Raspberry	2	1	20
Blueberry	2	3	18
Strawberry	1	1/2	8
Unit price	15	20	
Unit cost	10	16	

Georg, an employee at Waldgeist, has already modeled the above production problem as a linear program (P). By using the Simplex algorithm he found the following optimal table. Unfortunately, Georg was too conscientious in the quality control of Nachtblau and spilled some of the good wine on his invoice, which made some entries unreadable.

$$(P) \quad \underset{\text{s.t.}}{\text{Max}} \quad 5x_1 + 4x_2$$
$$\begin{aligned} 2x_1 + x_2 &\leq 20 \\ 2x_1 + 3x_2 &\leq 18 \\ x_1 + {}^1\!/_2 x_2 &\leq 8 \\ x &\in \mathbb{R}^2_{\geq 0} \end{aligned}$$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z			0	0.75	3.5	
			1	0	-2	
x_2			0	0.5	-1	
x_1			0	-0.25	1.5	7.5

a) Explain the meaning of the objective function, the constraints and the decision variables of (P)

b) Complete the final table above for Georg.

c) What is the maximum amount that Steffen should pay for $1kg$ of additional raspberries? Justify your answer.

d) Calculate the limits for the selling price of Nachtblau within which the optimal basis does not change. Explain in detail how the optimal production schedule and resource stocks change, if the selling price violates these limits.

e) The supply of strawberries and blueberries varies greatly.
   (i) Within what limits can the blueberry capacity move so that both products can continue to be produced?
   (ii) Within what limits can the strawberry capacity move so that both products can continue to be produced?
   (iii) Determine the respective shadow prices and reduced costs if the strawberry inventory violates these limits.

f) Steffen has the opportunity to exchange strawberries and blueberries for each other at the market. For q kilograms of one type of berry, he receives q kilograms of the other type of berry. Calculate the best split of strawberries and blueberries while maintaining the current base of the optimal table.

g) A new product *Fruchttraum* contains $4kg$ raspberries, $2kg$ blueberries and $3kg$ strawberries. Furthermore, Fruchttraum generates costs of 13 per bottle.
   (i) What is the minimum bottle price of Fruchttraum that must be in order for it to be included in the optimal production program?
   (ii) What happens if Fruchttraum needs $6kg$ of strawberries per bottle?

h) Steffen points out to Georg that only whole (and full!) bottles of Nachtblau and Himmelrot can be produced and therefore the solution calculated by Georg cannot be implemented.
   (i) How does this change the problem (P)?
   (ii) What does the solution calculated by Georg tell you about the achievable profit of the adapted problem?

i) *Self-study*: Solve the above subtasks using Gurobi.

Solution

a) x_1 Amount of Himmelrot produced
   x_2 Amount of Nachtblau Produced
   Obj profit maximization
   $Res1$: Do not use more than 20 units of raspberries for production
   $Res2$: Do not use more than 18 units of blueberries for production
   $Res3$: Do not use more than 8 units of strawberries for production
   ≥ 0 no negative production quantities allowed, otherwise arbitrarily real

b) First we fill in the base. We see that the base variable for the first constraint is x_3. Furthermore, x_1 is the base variable for the 3rd constraint and x_2 for the 2nd constraint. Since the shadow prices of x_4 and x_5 are positive, x_4 and x_5 must be 0. We thus get: $x_1 + 1/2 \cdot x_2 = 8$ and because $x_1 = 7.5$, $x_2 = 1$ must hold. We insert this into the first constraint and into the objective and get $x_3 = 4$ and $z = 41.5$.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	0	0	0	0.75	3.5	41.5
x_3	0	0	1	0	-2	4
x_2	0	1	0	0.5	-1	1
x_1	1	0	0	-0.25	1.5	7.5

c) The shadow price of an additional unit of raspberries is 0. (There are still 4 units of raspberries left.) Therefore, Steffen should not spend anything on additional raspberries.

d) We add Δ to the price of x_2 in the objective, which leads to the following in the final table:

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	0	$-\Delta$	0	0.75	3.5	41.5
Z	0	0	0	$3/4 + 1/2\Delta$	$7/2 - \Delta$	$41.5 + \Delta$

So that the basis does not change, all objective coefficients must be ≥ 0. So $3/4 + 1/2\Delta \geq 0$ and $7/2 - \Delta \geq 0$. This gives Δ: $-3/2 \leq \Delta \leq 7/2$ and therefore $p(x_2)$ must be $\in [18.5, 23.5]$.

When the price of Nachtblau falls below 18.5, the coefficient of x_4 becomes negative, x_4 enters the base with value 2 and kicks out x_2. This means Steffen would stop producing Nachtblau in favor of Himmelrot, of which 8 bottles are now produced. However, not all blueberries would be processed anymore (there is 2 left). There are still 4 raspberries left and strawberries are still a scarce commodity.

If the price of Nachtblau rises above 23.5, the coefficient of x_5 becomes negative, x_5 enters the base with value 5 and kicks out x_1. This means that Waldgeist would stop producing Himmelrot in favor of Nachtblau, of which 6 bottles are now produced. However, not all strawberries (5 left over) and raspberries (14 left over) would be processed. Blueberries remain a scarce commodity.

e) (i) The slack variable for blueberries is x_4. If we increase the blueberry inventory by Δ_B, we need to add Δ_B times the x_4 column to the right side. It must be ensured that the tableau remains feasible and that both products continue to be produced. We thus get the following conditions: $1 + 0.5\Delta_B > 0$ and $7.5 - 0.25\Delta_B > 0$, i.e. $\Delta_B \in (-2, 30)$. Thus, the blueberry inventory must remain in the interval $(16, 48)$ to guarantee that both products continue to be produced.

(ii) The slack variable for strawberries is x_5. If we increase the strawberry inventory by Δ_S, we need to add Δ_S times the x_5 column to the right side. It must be ensured that the tableau remains permissible and that both products continue to be produced. We thus get the following conditions:
$4 - 2\Delta_S \geq 0$ and $1 - \Delta_S > 0$ and $7.5 + 1.5\Delta_S > 0$, i.e. $\Delta_S \in (-5, 1)$. This means that the strawberry inventory must remain in the interval (3.9) to guarantee that both products continue to be produced.

(iii) We start with $\Delta_S = -5$. In this case, the right side of the 3rd constraint becomes 0, and thus x_1 is the exiting BV. We add t times the 3rd constraint to the objective function and evaluate which variable enters the basis:

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	t	0	0	$0.75 - 0.25t$	$3.5 + 1.5t$	–
x_1	1	0	0	-0.25	1.5	0
$Z(t = 3)$	3	0	0	0	8	–

Therefore, t would have to remain $\in [0, 3]$ for the tableau to remain optimal. For $t \geq 3$, x_4 enters the basis and the shadow prices and reduced costs change as shown in the tableau above.

Now let $\Delta_S = 1$. In this case, the right side of the 2nd constraint becomes 0, and thus x_2 is the exiting BV. We add t times the 2nd constraint to the objective function and evaluate which variable enters the basis:

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	0	t	0	$0.75 + 0.5t$	$3.5 - t$	–
x_2	0	1	0	0.5	-1	0
$Z(t = 3.5)$	0	3.5	0	2.5	0	–

Therefore, t would have to remain $\in [0, 3.5]$ for the tableau to remain optimal. For $t \geq 3.5$, x_5 enters the basis and the shadow prices and reduced costs change as shown in the tableau above.

f) If we increase the blueberry inventory by Δ units, we also decrease the strawberry inventory by Δ. This means we have to add Δ times the x_4 column to the right side and at the same time subtract Δ times the x_5 column. We obtain:

$$\begin{array}{lrcrl} (Z) & 41.5 + {}^{3}/_{4}\Delta - {}^{7}/_{2}\Delta & = & 41.5 - {}^{11}/_{4}\Delta & \rightarrow max \\ (I) & 4 + 2\Delta & & & \geq 0 \\ (II) & 1 + {}^{1}/_{2}\Delta + \Delta & = & 1 + {}^{3}/_{2}\Delta & \geq 0 \\ (III) & 7.5 - {}^{1}/_{4}\Delta - {}^{3}/_{2}\Delta & = & {}^{15}/_{2} - {}^{7}/_{4}\Delta & \geq 0 \end{array}$$

Thus we get: $\Delta \in [-{}^{2}/_{3}, {}^{30}/_{7}]$. So (Z) is maximized for $\Delta = -{}^{2}/_{3}$. Steffen should therefore exchange ${}^{2}/_{3}$kg of blueberries for the same amount of strawberries to increase his profit to $43\frac{1}{3}$.

g) Due to the loss of the ingredients required for the production of a Fruchttraum from the previous production plan, the objective function value is reduced.

(i) We want to calculate our imputed costs using shadow prices. If we make a bottle of Fruchttraum, our berry inventory will decrease by 4kg of raspberries, 2kg of blueberries and 3kg of strawberries. We need to check whether our current shadow prices are still valid for these changes. We know from subtask e(i) that the shadow price for blueberries remains valid if our blueberry inventory does not decrease by more than 2kg. In addition, according to subtask e(ii), the shadow price for strawberries also remains valid if the strawberry inventory does not decrease by more than 5kg. From our optimal table we also see that with the current production plan there are currently 4kg of raspberries left. In this respect, the shadow price for raspberries remains the same if the stock does not decrease by more than 4kg. So we can calculate our imputed costs as follows: $4 \cdot 0 + 2 \cdot \frac{3}{4} + 3 \cdot \frac{7}{2} = 12$. After that, a bottle of Fruchttraum should cost at least $12 + 13 = 25$.

(ii) As shown in exercise e(ii), the shadow price of strawberries changes when more than $5kg$ of strawberries are withdrawn from production. For the last of the $6kg$ strawberries, the new shadow price must be taken into account, which was already calculated in subtask e(iii): $4 \cdot 0 + 2 \cdot \frac{3}{4} + 5 \cdot \frac{7}{2} + 1 \cdot 8 = 27$. So a bottle of Fruchttraum should cost at least $27 + 13 = 40$.

At this point, for the sake of completeness, you would have to check whether the new strawberry shadow price of 8 is also valid for at least $1kg$. In problem part e(iii) we have already determined that x_1 leaves the base and x_4 enters the base when the strawberry population violates the lower bound. So we do this base swap, restore the canonical form, and then use the right-hand sides again to calculate the scope of the new strawberry shadow price.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	0	0	0	0.75	3.5	$41.5 + 3.5\Delta_S$
x_3	0	0	1	0	-2	$4 - 2\Delta_S$
x_2	0	1	0	0.5	-1	$1 - \Delta_S$
x_1	1	0	0	-0.25	1.5	$7.5 + 1.5\Delta_S$

Base exchange $x_1 \leftrightarrow x_4$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	3	0	0	0	8	$64 + 8\Delta_S$
x_3	0	0	1	0	-2	$4 - 2\Delta_S$
x_2	2	1	0	0	2	$16 + 2\Delta_S$
x_4	-4	0	0	1	-6	$-30 - 6\Delta_S$

In order for the tableau to remain feasible, the following inequalities must hold: $4 - 2\Delta_S \geq 0$, $16 + 2\Delta_S \geq 0$ and $-30 - 6\Delta_S \geq 0$. This results in $\Delta_S \in [-8, -5]$. The new strawberry shadow price of 8 is therefore always valid when the strawberry inventory is in the interval [0.3]. Since we have a total of $8kg$ of strawberries at our disposal, but we need $6kg$ for the new product, we can safely use the new shadow price for the 6th kilogram of strawberries.
Please note that in the above calculation it would not have been necessary to recalculate the entire tableau. Instead, it would have been sufficient to just recalculate the right side.

h) (i) $x \in \mathbb{R}^2_{\geq 0} \longrightarrow x \in \mathbb{N}^2_0$

(ii) Determined profit is upper bound for the profit of the integer problem